

ligent devices that can enable a sustainable planet and improve the lives of both current and future generations. This is the biggest return on investment from green products which very much justifies the cost of going green," says Ronnie Ho, general manager, microcontrollers business unit, Renesas Electronics Singapore.

Focus areas

The solution lies in product design. The product has to be designed for environment. Basically, there are four things to consider: The product should not have any toxic chemical. It should have a longer life than the average life of the product in the market (for instance, in the case of laptops, it is five years). It should be easily upgradable and completely recyclable. It should also be highly energy-efficient as per the prescribed industry standard.

"Every design and development can be green. The differentiator is "how much green." There are limitations and challenges to embedded development. You need a certain power to turn on a fan and you also need a certain power to run the engine of a car. Both the applications demand 'green' design," says Ronnie Ho.



Electronic waste in landfills has been found to contain high levels of harmful chemicals (Image courtesy: www.greenpeace.org)

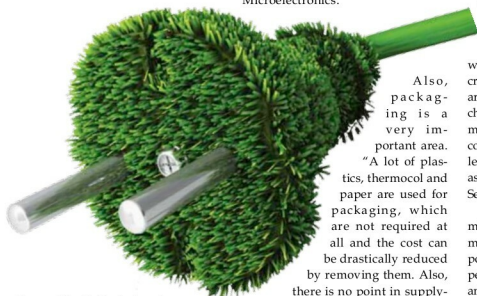
"The adoption of RoHS-compliant components in phased manner can be cost-effective. RoHS-compliant components are now easily available in India. Replacement of the old bill of material with new RoHS-compliant materials provides a smooth transition. The vendor of electronic components can provide direct replacement of the legacy products," suggests Manoj Kumar, senior group manager, IMS (industrial and multisegment sector), ST-Microelectronics.

started giving CDs. Nokia has for the first time started the practice of providing all the operating instructions within the product itself. The operating manual comes up as you switch on the phone," shares Pratap.

Not just the cabling and packaging, one has to look at the entire product design. Using an LED screen rather than an LCD screen saves a lot of money because LED screens don't have any mercury content.

"Basically, the demand for green products comes from end customers' requirement of cutting their energy costs. Mostly all the consumer sectors now demand products which are power savers. With the increasing carbon footprint, consumers are becoming more conscious of their choices of the products. Semiconductor manufacturers are also taking care to come out with devices that consume lesser power," says Vinay Kumar, assistant marketing manager, Fujitsu Semiconductor Asia.

"For instance, we have dedicated microcontrollers for brushless DC motor applications which reduce power consumption by around 60 per cent. Apart from this, we have an ecology memory product called FRAM which doesn't need any battery for functioning. FRAM can



Energy-efficient lighting is also a focus area (Image courtesy: www.greenlightenergyservices.com)

Also, packaging is a very important area. "A lot of plastics, thermocol and paper are used for packaging, which are not required at all and the cost can be drastically reduced by removing them. Also, there is no point in supplying paper menus with digital equipment. Many companies have